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A COMPARATIVE ANALYSIS OF AGILE METHODOLOGY

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A Comparative Analysis of Agile Methodology

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Abstract— Agile methodology is a dynamic process where a number of cycles are run to deliver a product with a higher customer value. The purpose of this paper is to discuss the benefits and drawbacks of using Agile in contrast to other frameworks such as Waterfall and determine the principal distinctions between them. By doing this comparison, the paper is able to determine circumstances under which Agile is most efficient and circumstances in which Agile may experience some issues. The research shows that due to Agile's flexibility it is an effective method in environments where changes are frequent, however, due to the constant involvement of the customer and the relative difficulty in applying Agile at a large scale, it may not be suitable for large organizations.

Keywords— Agile methodology, software development, Waterfall, Scrum, Kanban, project management, iterative development.

I. INTRODUCTION

Agile methodology's flexibility allows it to adapt effectively in environments where changes are frequent, making it an optimal choice for projects that demand quick iterations and frequent feedback [2]. Originally developed in 2001 as the Agile Manifesto, Agile is now the go-to process for many organizations that require fast response to the customer's requirements. Unlike traditional models of software development, Agile divides development work into smaller, more manageable segments that can be released with functional software and respond quickly to feedback. This methodology aligns with frameworks like Scrum and Kanban which are used in both software and the non-software industries.

The most basic and traditional approach is the Waterfall model where one phase has to be done before the next phase is begun. This model is designed with a definite structure and is predictable in nature, it is therefore appropriate for a project with clearly defined specifications. However, Waterfall lacks flexibility in responding to the changing customer requirements as it has a linear model that prevents a return to any completed phase. The output of a project, like software development, frequently falls short of the customer's expectations because of a rigid approach to achieving the goals, which leaves little room for adaptability to changing client demands.

Due to the pressure of declining the time taken in the market, and the satisfaction of the customers, Agile has found much popularity among businesses. The daily communication and the numerous cycles embedded in Agility provide the chance for fast changes which is one of the reasons why it is used. The objective of this paper is to contrast Agile with other methods, including Waterfall, discuss the strengths and weaknesses of Agile, and evaluate the effectiveness of the method in actual practice.

II. COMPARATIVE ANALYSIS

There is a clear distinction between the Agile and the Waterfall methodologies of project management. Waterfall is a linear model where the process is divided into phases and all of them can only begin after the previous one is done. This structure makes it very appropriate for projects that have clear, stable and non-changing requirements. This is good because it gives a clear direction and expectation on the team, few change orders are issued. However, it lacks the flexibility which is required when a phase is completed, it is difficult to retrace the previous steps, and again it becomes a problem when change occurs later in the development process.

While Agile is more flexible and focuses on the project's development in stages. The work in agile projects is broken down into short time frames, called "sprints," during which the project is reviewed and reprioritized according to customer feedback or the needs of the project as it progresses [2]. This procedure enables change and frequent delivery of functional software, which makes the project meet customer needs as it is being developed. This makes Agile especially important in those situations where requirements are sure to change, for instance in the software development industries. This is illustrated by Scrums, one of the most popular Agile frameworks which subdivides all the tasks into smaller parts.

The flexibility that Agile offers is its biggest advantage, but it also has its own problems. Customer interaction is integrated throughout the Agile process and communication occurs frequently, which can be problematic if stakeholders are not easily accessible or if there is limited direction. This increased dependency on collaboration can at times hinder the speed of working or result in

miscommunication especially where the team members are scattered in their working stations or the team operates across different time zones. Furthermore, the short-term orientation of Agile may, despite a better individual project execution make overall planning and strategic alignment more challenging, since the target is not in view all the time.

Although Agile's adaptability is its greatest advantage, this methodology can present challenges in large-scale projects, where the need for synchronized collaboration across teams becomes complex [3]. Supervising two or more groups who are working in different phases but are dependent on one another, need to be well coordinated to avoid duplication of work. There might be a better way to integrate Agile with the organizations that cannot go all in with the flexible approach, such as Large-scale projects may be better served by hybrid models that incorporate Agile with more rigid methodologies such as Waterfall. Because these are hybrids, feedback can occur in a loop while still adhering to a framework to guarantee that overall objectives are achieved. That is why, although Agile is much more flexible in comparison with the traditional methodologies, it is effective on the condition that it is used in a certain environment.

III. OPINIONS

As businesses navigate constant change, Agile practices enhance flexibility, promote teamwork, and generate products centered around customer needs. Due to this, Agile methodologies have gained much more popularity in the system development life cycle (SDLC). Many big companies and SMEs across various industries have switched to this approach to streamline their development process, such as Google, Microsoft and Amazon.

The rising popularity and effectiveness of Agile is evident in its increasing adoption. Digital.ai reports that Agile usage saw a twofold increase within just a year. In 2020, 37% of companies had experimented with Agile at least once, but by 2021, this percentage rose sharply to 78%, including even brief users.

From this data source, it is quite evident that Agile has become the next development methodologies trend as proven by the increased usage of Agile method. However this also brings various alternatives of Agile methods to better suit their business model. Currently, there are over ten widely used Agile methodologies, and while they all share much of the same overarching philosophy, as well as many of the same characteristics and practices, there are still some differences in their terminology, practices and operational tactics.

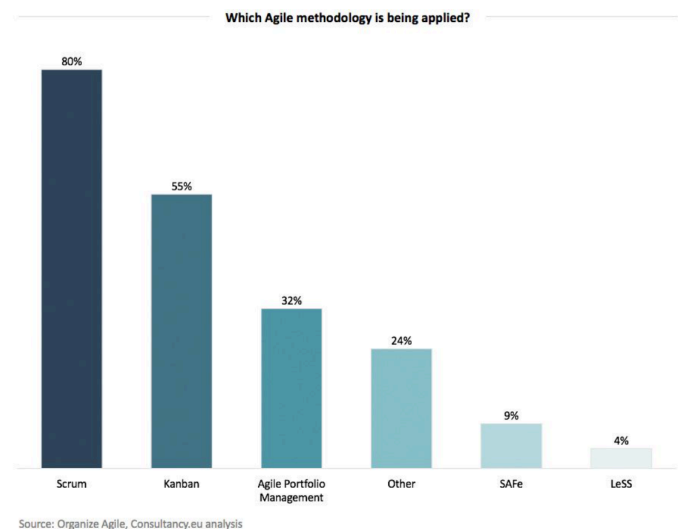


Figure 1 Most Used Agile Methodology

IV. ADVANTAGES AND DISADVANTAGES

In this section we will delve deeper to identify what makes the Agile method stand out among the other methodologies and become the preferred approach in SDLC. Below are the benefits.

- **Satisfied clients**
As stakeholders are involved in a system development life cycle, the outcome of the project will leave them satisfied as they are always engaged and giving feedback on the system requirements specifications.
- **Improved quality**
As Agile involves an iterative approach process, developers are able to review and further improve the current system as each interval is repeated.
- **Adaptability**
Agile method is highly responsive to changes due to its iterative approach, hence allowing for an easier way to deliver consistently and manage client's new requests effectively.
- **Predictability**
Companies use estimates for feasibility analysis and project planning. Due to the short time period in each iteration, it is much simpler for managers to handle resources and split the tasks among the team. Not to mention it allows better judgement on the cost prediction for a shorter term.
- **Reduced risk**
Developers frequently assess progress during each sprint, making it easier for them to detect any potential issues in the project and hence creating a better system in the long run.

- Team communication

Agile promotes frequent collaboration and communication between team members to ensure everyone is on the same track and to keep check of their progress.

Though on the surface Agile seems to bring a lot of benefits, it also has its own weaknesses. Companies implementing Agile should also consider the weaknesses of the Agile model. The most important of these are listed below.

- Lack of documentation

Tasks are often finished just in time for development, documentation are far less detailed and thus may cause confusion and misunderstanding of the project in the long term.

- Scope creep

Client's requirements change regularly, leading to widening of projects' scope and frequent confusion in which requirements to prioritize first.

- Time constraint

During the sprint process team members are forced to continuously perform in engagement with the stakeholders and also involved in the development of the project, hence likely to cause fatigue and burnout to the team members.

- Complex regulatory

Long-term projects, particularly in sectors like healthcare, finance, or government, often have strict regulatory and compliance requirements that need upfront planning and documentation. Agile's iterative, evolving approach may not suit environments that require detailed, predetermined documentation from the start.

V. CONCLUSIONS

What makes Agile stand out from the Waterfall method is its flexibility in adapting to changing requirements throughout the development process rather than a linear approach where backtracking to earlier stages is mostly impossible. Not to mention it also underscores the highly-demanding constant interaction and collaboration between developers and the stakeholders, creating a system that satisfies their needs.

Overall, Agile fosters a dynamic and responsive development environment, making it particularly effective in fast-paced industries where requirements may evolve during the project lifecycle. Incorporating this model into the project life cycle has become imperative for both large corporations and small to medium-sized enterprises, as it offers a transformative approach that drives efficiency, innovation, and long-term success.

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